A New Framework of LSB Steganalysis of Digital Media

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***Abstract*—We propose a general framework for the detection of the least significant bit (LSB) steganography using digital media files as cover objects. The new framework exploits high-order statistics of the samples. It can compute a robust estimate of the length of a secret message hidden in the LSBs of samples for a large class of digital media contents such as image, video, and audio, in which the underlying signals consist of correlated samples. A case study on the LSB steganalysis of natural grey-scale and color images and experimental results are reported.**

***Index Terms*—Digital media, LSB steganography, steganalysis.**

I. Introduction

THE profusion of digital media (image, video, and audio) in our modern life has led to a rapid technological development of steganography and steganalysis with digital media files being the carrier contents (camouflage). A widely used digital steganography technique is the so-called least significant bit embedding (LSB embedding). In LSB steganography, the secrete messages, probably encrypted, are hidden in the LSBs of the samples of a digital signal. The perceptual transparency of the LSB steganography is easily achieved by the camouflage of noises inherent to digital signal acquisition process. In designing digital sensors, the quantizer sets the LSB of the binary sample representation at or even below the level of sensor noise energy in order not to lose precision. This common engineering practice has seemingly made LSB steganography an attractive and popular technique.

In recent years, a number of papers were published on steganalysis of LSB embedding in grey-scale and color images [1]–[7], [9]–[11]. These authors aimed to estimate the length of the embedded message, if any, by exploiting the sample correlations in natural image signals. In [5], Fridrich *et al.* proposed a fairly general detection methodology. It involves in identifying a function $S(\ell)$ (called distinguishing statistics) that predictably changes with the length ℓ of the embedded message. The function $S(\ell)$ is determined by estimating its parameters (coefficients) from some known extreme values of S, such as $S(0)$ (corresponding to the clean signal) and $S(\ell_{\max})$ (the value of S if all the signal samples are used in embedding). Finally, the message length ℓ is computed by solving the equation $S(\ell) = S'$, where S' is the value of S for the signal (potentially stego signal) under examination. This approach requires solving an equation, but the the coefficients of the equation need to be estimated as part of the detection algorithm.

We also proposed an LSB steganalysis technique called sample pair analysis that estimates the hidden message length by solving a quadratic equation [2]. From the signal being steganalyzed, the coefficients of our equation can be computed using closed analytical formulae, rather than empirically determined by adding embedding noises to the signal being tested. This results in a faster detection algorithm. Very recently, a variant of the technique of [2] was proposed by Lu *et al.* They treated the problem as one of least-square estimation and reported improved estimation accuracy on a set of test images [10].

Our earlier technique of [2] uses the statistics of pairs of signal samples. In the present paper, we attempt to derive detection equations based on the statistics of larger groups of correlated samples, hence, utilizing higher order statistics. This is motivated by the intuition that more robust estimation should be made possible by exploiting the higher order correlation of signal samples.

Perhaps more significantly, this paper sheds some light on and presents a general framework for a class of LSB steganalysis techniques that estimate hidden message length by measuring some signature statistical quantity (distinguishing statistics in the terminology of [5]). Specifically, the signature statistics is given as a function $S(\ell)$ of the hidden message length ℓ, but instead of solving the equation $S(\ell) = S'$, we develop an equivalent equation with known coefficients. The main idea is to construct a signature statistics $S(\ell)$ as a function F (with known coefficients) of some feature vector $\mathbf{V}(\ell)$ of the signal, which is sensitive to the length ℓ of the hidden message, namely

$$S(\ell) = F(\mathbf{V}(\ell)). \tag{1}$$

Although the value $\mathbf{V}(0)$ of this vector for the clean signal is not known to the steganalyst, the value $\mathbf{V}(\ell)$ for the signal being steganalyzed can be measured. We will express $\mathbf{V}(0)$ as a function G (also with known coefficients) of $\mathbf{V}(\ell)$ and ℓ: $\mathbf{V}(0) = G(\mathbf{V}(\ell), \ell)$. Thus, the relation $F(\mathbf{V}(0)) = S(0)$ becomes equivalent to $F(G(\mathbf{V}(\ell), \ell)) = S(0)$. A key issue is to choose S such that $S(0)$ is approximately a constant for the class of signals being examined. If $S(0)$ is a constant, then the above equation in ℓ has all its coefficients determined by the given signal being tested; hence, the message length ℓ can be estimated.

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The paper presentation is organized as follows. In Section II, we lay down a general framework for LSB steganalysis that exploits high order sample statistics. Section III introduces a feature vector that aids steganalysis and reveals its properties. In Section IV, we derive a characteristic function with the feature vector being its argument. This characteristic function leads to polynomial equations in the hidden message length, which can be used to detect LSB steganography. To present a concrete example of using the proposed steganalysis framework in the real world, we conduct, in Section V, a case study for steganalysis of LSB embedding in continuous-tone images. Empirical evidence is presented to demonstrate the efficacy of the proposed steganalysis technique. Section VI concludes the paper.

II. General Detection Framework

Pivotal to the development and performance of our new framework for LSB steganalysis is the selection of a feature vector $\mathbf{V}$ that is highly sensitive to the length ℓ of the hidden message. Throughout this paper, the length ℓ of the hidden message should be understood as a quantity relative to the length of the signal sequence being tested. For instance, in the case of image steganalysis, let r be the percentage of the pixels in a given image whose LSBs carry the hidden message bit. Then, we define $\ell \equiv r/2$ for a convenience to be self-evident shortly. In other words, for the case of random embedding, ℓ is nothing but the probability that the LSB of a sample will be flipped by an LSB stegnography algorithm.

There are two criteria for selecting the feature vector $\mathbf{V}$. Besides $\mathbf{V}(\ell)$ being sensitive to ℓ, the second criterion is that $\mathbf{V}(0)$ can be expressed in terms of $\mathbf{V}(\ell)$ and ℓ. In our quest for the desired feature vector $\mathbf{V}$, we start with finding a function G such that $\mathbf{V}(0) = G(\mathbf{V}(\ell), \ell)$. In addition, in order to exploit the higher order correlation of the signal, we construct $\mathbf{V}$ based on the kth-order statistics of the signal.

Let $\mathbf{X} = (X_1, X_2, \ldots, X_k)$ be the random vector of k samples in a clean signal. The spatial disposition of the samples in $\mathbf{X}$ needs to be carefully chosen such that the samples are highly correlated. For typical digital media signals such as image, video, and audio, the samples should be spatially adjacent. In other words, $\mathbf{X}$ is formed of a contiguous sample block.

Assume that the binary representation of sample values uses B bits. Let the random binary vector $\mathbf{U} = (U_1, U_2, \ldots, U_k)$ be the vector of the LSBs of $\mathbf{X} = (X_1, X_2, \ldots, X_k)$, and accordingly, the random vector $\mathbf{Y} = (Y_1, Y_2, \ldots, Y_k)$ represents the values of $\mathbf{X} = (X_1, X_2, \ldots, X_k)$ after the LSB bits are truncated. Hence, $X_i = 2Y_i + U_i$ for all i. Let $\mathbf{X}' = (X_1', X_2', \ldots, X_k')$, $\mathbf{U}' = (U_1', U_2', \ldots, U_k')$, and $\mathbf{Y}' = (Y_1', Y_2', \ldots, Y_k')$ denote the counterparts for the signal altered by random LSB embedding.

The process of LSB embedding does not corrupt the most significant $B-1$ bits of the samples, and thus, the random vector $\mathbf{Y}$ remains unchanged: $\mathbf{Y}' = \mathbf{Y}$. On the other hand, since the hidden message bits are randomly scattered in time/space, any bit in the sequence of LSBs has an equal chance to be flipped, independent of the other bits in the sequence. Therefore, the event that a binary sequence $\mathbf{v}$ is changed by LSB steganography to another binary sequence $\mathbf{u}$, both having k bits, has the probability

$$P(\mathbf{U}' = \mathbf{u} | \mathbf{U} = \mathbf{v}) = \ell^{d(\mathbf{u},\mathbf{v})}(1-\ell)^{k-d(\mathbf{u},\mathbf{v})} \tag{2}$$

where $d(\mathbf{u}, \mathbf{v})$ is the Hamming distance between the two binary vectors $\mathbf{u}$ and $\mathbf{v}$.

Clearly, the distribution of the random vector $\mathbf{U}'$ can be expressed in terms of the distribution of $\mathbf{U}$ and the probability ℓ of flipping a bit, and conversely, the distribution of $\mathbf{U}$ can be expressed as a function of the distribution of $\mathbf{U}'$ and the flipping probability ℓ. However, the probability mass function (pmf) of $\mathbf{U}'$ is not a good choice of the feature vector $\mathbf{V}$ for steganalysis because the pmf of $\mathbf{U}$ is uniform for signals encountered in digital media. As a result, the pmf of $\mathbf{U}'$ is also uniform and, hence, insensitive to LSB embedding.

Instead, we consider as the feature vector $\mathbf{V}$ the joint pmf of $\mathbf{U}$ and the random variable $f(\mathbf{Y})$, where f is a properly chosen real-valued function defined on k-dimensional real vectors. Since the LSB embedding does not affect the most significant $B-1$ bits, or $\mathbf{Y}' = \mathbf{Y}$, we have

$$\begin{aligned} &P(f(\mathbf{Y}') = m, \mathbf{U}' = \mathbf{u}) \\ &\quad = \sum_{\mathbf{v} \in \{0,1\}^k} P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{v}) P(\mathbf{U}' = \mathbf{u} | \mathbf{U} = \mathbf{v}). \end{aligned} \tag{3}$$

It is convenient to write the above in a matrix form. Denote by $\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{2^k-1}$ all possible k-bit binary sequences. Let $\mathbf{Q}(\ell)$ be the steganography transition matrix with elements $Q_{i,j}(\ell) = P(\mathbf{U}' = \mathbf{u}_i | \mathbf{U} = \mathbf{u}_j)$, $0 \le i, j \le 2^k - 1$. Consequently

$$Q_{i,j}(\ell) = \ell^{d(\mathbf{u}_i,\mathbf{u}_j)}(1-\ell)^{k-d(\mathbf{u}_i,\mathbf{u}_j)}. \tag{4}$$

Then

$$\begin{aligned} &\begin{pmatrix} P(f(\mathbf{Y}') = m, \mathbf{U}' = \mathbf{u}_0) \\ P(f(\mathbf{Y}') = m, \mathbf{U}' = \mathbf{u}_1) \\ \vdots \\ P(f(\mathbf{Y}') = m, \mathbf{U}' = \mathbf{u}_{2^k-1}) \end{pmatrix} \\ &\qquad = \mathbf{Q}(\ell) \begin{pmatrix} P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u}_0) \\ P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u}_1) \\ \vdots \\ P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u}_{2^k-1}) \end{pmatrix}. \end{aligned} \tag{5}$$

As proved in Appendix A, for $\ell \in [0, 0.5)$, the matrix $\mathbf{Q}(\ell)$ is invertible with inverse $\mathbf{Q}^{-1}(\ell)$; hence

$$\begin{aligned} &\begin{pmatrix} P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u}_0) \\ P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u}_1) \\ \vdots \\ P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u}_{2^k-1}) \end{pmatrix} \\ &\qquad = \mathbf{Q}^{-1}(\ell) \begin{pmatrix} P(f(\mathbf{Y}') = m, \mathbf{U}' = \mathbf{u}_0) \\ P(f(\mathbf{Y}') = m, \mathbf{U}' = \mathbf{u}_1) \\ \vdots \\ P(f(\mathbf{Y}') = m, \mathbf{U}' = \mathbf{U}_{2^k-1}) \end{pmatrix}. \end{aligned} \tag{6}$$

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Consider the feature vector $\mathbf{V}(\ell)$ to be the $M \times 2^k$-dimensional vector whose components are $P\left(f(\mathbf{Y}') = m, \mathbf{U}' = \mathbf{u}_i\right)$ for all m and $0 \le i \le 2^k - 1$, where M is the total number of possible values of m. Let $\bar{\mathbf{Q}}(\ell)$ denote the $M \times 2^k$-dimensional matrix

$$\bar{\mathbf{Q}}(\ell) = \begin{pmatrix} \mathbf{Q}(\ell) & 0 & \cdots & 0 \\ 0 & \mathbf{Q}(\ell) & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \mathbf{Q}(\ell) \end{pmatrix}. \tag{7}$$

Equations (5) and (6) imply, respectively, that

$$\mathbf{V}(\ell) = \bar{\mathbf{Q}}(\ell)\mathbf{V}(0) \tag{8}$$

and

$$\mathbf{V}(0) = \bar{\mathbf{Q}}^{-1}(\ell)\mathbf{V}(\ell). \tag{9}$$

As we have explained in the introduction, in order to derive the detection equation, we need to construct a real-valued function $F(\mathbf{V})$ of the feature vector $\mathbf{V}$ such that $F(\mathbf{V}(0)) = F_0$ for some signal-independent constant F_0. Furthermore, we want $F(\mathbf{V}(\ell)) \neq F_0$ for all $\ell \in (0, 0.5)$. We will call $F(\mathbf{V})$ the characteristic function. Then, the detection equation becomes

$$F(\bar{\mathbf{Q}}^{-1}(\ell)\mathbf{V}') = F_0 \tag{10}$$

where $\mathbf{V}'$ is the value of the feature vector $\mathbf{V}$ for the tested signal. Indeed, a solution of (10) is the probability ℓ_0 that an LSB bit is flipped in the tested signal because $\mathbf{V}(0) = \bar{\mathbf{Q}}^{-1}(\ell_0)\mathbf{V}'$, and $F(\mathbf{V}(0)) = F_0$. Moreover, we show next that ℓ_0 is the smallest non-negative solution of (10). For this, it suffices to show that $F(\mathbf{V}'') \neq F_0$ for any $\ell \in [0, \ell_0)$, where $\mathbf{V}'' = \bar{\mathbf{Q}}^{-1}(\ell)\mathbf{V}'$. Note that since $\mathbf{V}' = \bar{\mathbf{Q}}(\ell_0)\mathbf{V}(0)$, it follows that

$$\mathbf{V}'' = \bar{\mathbf{Q}}^{-1}(\ell)\bar{\mathbf{Q}}(\ell_0)\mathbf{V}(0). \tag{11}$$

In Appendix A, the following equality is proved.

$$\bar{\mathbf{Q}}(\ell_1)\bar{\mathbf{Q}}(\ell_2) = \bar{\mathbf{Q}}(\ell_1 + \ell_2 - 2\ell_1\ell_2). \tag{12}$$

Let $\ell_2 = (\ell_0 - \ell/1 - 2\ell)$. Then, $\ell + \ell_2 - 2\ell\ell_2 = \ell_0$. Moreover, since $0 \le \ell < \ell_0 \le 0.5$, we have $0 < \ell_2 \le \ell_0$. Applying (12) yields

$$\bar{\mathbf{Q}}(\ell)\bar{\mathbf{Q}}(\ell_2) = \bar{\mathbf{Q}}(\ell_0) \tag{13}$$

and further

$$\bar{\mathbf{Q}}^{-1}(\ell)\bar{\mathbf{Q}}(\ell_0) = \bar{\mathbf{Q}}(\ell_2). \tag{14}$$

By substituting (14) into (11), we obtain

$$\mathbf{V}'' = \bar{\mathbf{Q}}(\ell_2)\mathbf{V}(0). \tag{15}$$

The above relation implies that $\mathbf{V}'' = \mathbf{V}(\ell_2)$. Since $0 < \ell_2 < 0.5$ and by the property of the characteristic function that $F(\mathbf{V}(\ell_2)) \neq F_0$, we have $F(\mathbf{V}'') \neq F_0$, proving our claim.

III. Vector of Discriminating Features

Having showed in the previous section how to estimate the embedded message length ℓ in the proposed framework of LSB steganalysis, using an $M \times 2^k$-dimensional feature vector $\mathbf{V}(\ell)$ whose components are of form $P\left(f(\mathbf{Y}') = m, \mathbf{U}' = \mathbf{u}_i\right)$, $0 \le i \le 2^k - 1$, we proceed to find an appropriate function f to define $\mathbf{V}(\ell)$.

Note first that the sum of the elements on each row of matrix $\bar{\mathbf{Q}}(\ell)$ is 1. By (8), a feature vector $\mathbf{V}(0)$ of equal components implies $\mathbf{V}(\ell) = \mathbf{V}(0)$, but we want the feature vector $\mathbf{V}(\ell)$ to change with ℓ; thus, the components of $\mathbf{V}(0)$ have to differ from each other. Since the joint pmf of random vectors $f(\mathbf{Y})$ and $\mathbf{U}$ relates to the pmf of $f(\mathbf{X})$ via $\mathbf{X} = 2\mathbf{Y} + \mathbf{U}$, we seek to make the random variable $f(\mathbf{X})$ as biased as possible so that the components of $\mathbf{V}(\ell)$ can have significantly different values.

In search of a biased $f(\mathbf{X})$, we use the fact that the signals encountered in digital media are typically lowpass. Thus, we should make $f(\mathbf{X})$ a highpass filter of $\mathbf{X}$. Specifically, define f by

$$\begin{aligned} &f(x_1, x_2, \ldots, x_k) \\ &\quad = k\left(x_t - \frac{x_1 + \cdots + x_k}{k}\right) \\ &\quad = (k-1)x_t - (x_1 + \cdots + x_{t-1} + x_{t+1} + \cdots + x_k) \end{aligned} \tag{16}$$

where the position index $t = \lfloor k/2 \rfloor + 1$ labels the midpoint of a sequence. If $\mathbf{X}$ represents a lowpass waveform (e.g., a contiguous sample block in an image), then the random variable $f(\mathbf{X})$ has a generalized Gaussian (most likely Laplacian) distribution with 0 mean. Because this distribution is very biased, we may assume that there is an invertible linear transform $\mathbf{X} \to T(\mathbf{X}) = (f(\mathbf{X}), f_2(\mathbf{X}), f_3(\mathbf{X}) \ldots, f_k(\mathbf{X}))$, such that the random variable $f(\mathbf{X})$ and the random vector $T_1(\mathbf{X}) = (f_2(\mathbf{X}), f_3(\mathbf{X}) \ldots, f_k(\mathbf{X}))$ are decorrelated.

We are concerned with characterizing the joint distribution of $f(\mathbf{Y})$ and $\mathbf{U}$. Given an integer m and a k-bit binary sequence $\mathbf{u} = (u_1, u_2, \ldots, u_k)$, we have

$$\begin{aligned} P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u}) &= \sum_{f(\mathbf{y})=m} P(\mathbf{Y} = \mathbf{y}, \mathbf{U} = \mathbf{u}) \\ &= \sum_{f(\mathbf{y})=m} P(\mathbf{X} = (2\mathbf{y} + \mathbf{u}) \\ &= \sum_{f(\mathbf{y})=m} P(T(\mathbf{X}) = T(2\mathbf{y} + \mathbf{u})) \end{aligned} \tag{17}$$

where $\mathbf{y} \in \mathbb{Z}^k$, where $\mathbb{Z}$ is the set of all integers. Since $f(\mathbf{X})$ and $T_1(\mathbf{X})$ are decorrelated, we may assume that they are approximately independent and substitute $P(T(\mathbf{X})) = P(f(\mathbf{X}))P(T_1(\mathbf{X}))$ into (17), yielding

$$\begin{aligned} P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u}) = &\sum_{f(\mathbf{y})=m} P(f(\mathbf{X}) = f(2\mathbf{y} + \mathbf{u})) \\ &\times P(T_1(\mathbf{X}) = T_1(2\mathbf{y} + \mathbf{u})). \end{aligned} \tag{18}$$

Since f_i are linear functions, it follows that

$$f_i(2\mathbf{y} + \mathbf{u}) = 2f_i(\mathbf{y}) + f_i(\mathbf{u}). \tag{19}$$

In addition, for all $\mathbf{y}$, such that $f(\mathbf{y}) = m$, we have

$$f(2\mathbf{y} + \mathbf{u}) = 2f(\mathbf{y}) + f(\mathbf{u}) = 2m + f(\mathbf{u}). \tag{20}$$

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Consequently, (18) becomes

$$\begin{aligned} P(f(\mathbf{Y}) &= m, \mathbf{U} = \mathbf{u}) = \\ &\sum_{f(\mathbf{y})=m} P(f(\mathbf{X}) = 2m + f(\mathbf{u}))P(T_1(\mathbf{X}) \\ &= T_1(2\mathbf{y} + \mathbf{u}))P(f(\mathbf{X}) = 2m + f(\mathbf{u})) \\ &\sum_{f(\mathbf{y})=m} P(T_1(\mathbf{X}) = T_1(2\mathbf{y} + \mathbf{u})). \end{aligned} \tag{21}$$

Relation (19) implies that

$$T_1(2\mathbf{y} + \mathbf{u}) = (2f_2(\mathbf{y}) + f_2(\mathbf{u}), \ldots, 2f_k(\mathbf{y}) + f_k(\mathbf{u})). \tag{22}$$

Since T is an invertible transform, we perform a change of variable $\mathbf{y} \to \mathbf{z}$ such that $\mathbf{z} = T(\mathbf{y})$, in the summation of (21). Over all k-tuples of integers $\mathbf{y} \in \mathbb{Z}^k$, the vector $\mathbf{z}$ takes value in a lattice $\mathcal{L}$ in the k-dimensional space. Thus, we have

$$\begin{aligned} P(f(\mathbf{Y}) &= m, \mathbf{U} = \mathbf{u}) \\ &= P(f(\mathbf{X}) = 2m + f(\mathbf{u})) \\ &\sum_{\mathbf{z}\in\mathcal{L}, z_1=m} \\ P(T_1(\mathbf{X}) &= (2z_2 + f_2(\mathbf{u}), \ldots, 2z_k + f_k(\mathbf{u})). \end{aligned} \tag{23}$$

When $\mathbf{z} = (z_1, z_2 \ldots, z_k) \in \mathcal{L}$ and $z_1 = m$, the vector $(z_2, \ldots, z_k)$ takes values in a lattice $\mathcal{L}_m$ in the $(k-1)$-dimensional space. Note that for $m \neq m'$, $\mathcal{L}_{m'}$ is the image by a translation of $\mathcal{L}_m$. Moreover, when $(z_2, \ldots, z_k) \in \mathcal{L}_m$, then the vector $(2z_2 + f_2(\mathbf{u}), \ldots, z_k + f_k(\mathbf{u}))$ takes values in another lattice in the $(k-1)$-dimensional space, which we denote by $\mathcal{L}_{m,\mathbf{u}}$. In addition, for $m \neq m'$ and $\mathbf{u} \neq \mathbf{u}'$, the lattices $\mathcal{L}_{m,\mathbf{u}}$ and $\mathcal{L}_{m',\mathbf{u}'}$ are translations of one another. Therefore

$$\begin{aligned} &\sum_{\mathbf{z}\in\mathcal{L}, z_1=m} P(T_1(\mathbf{X}) = (2z_2 + f_2(\mathbf{u}), \ldots, 2z_k + f_k(\mathbf{u}))) \\ &= \sum_{(z_2,\ldots,z_k)\in\mathcal{L}_m} P(T_1(\mathbf{X}) = (2z_2 + f_2(\mathbf{u}), \ldots, 2z_k + f_k(\mathbf{u}))) \\ &= P(T_1(\mathbf{X}) \in \mathcal{L}_{m,\mathbf{u}}). \end{aligned} \tag{24}$$

Consequently

$$\begin{aligned} &P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u}) \\ &\quad = P(f(\mathbf{X}) = 2m + f(\mathbf{u}))P(T_1(\mathbf{X}) \in \mathcal{L}_{m,\mathbf{u}}). \end{aligned} \tag{25}$$

We are interested in how the value $P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u})$ varies with m and $\mathbf{u}$. Because the distribution of $f(\mathbf{X})$ is very biased, the factor $P(f(\mathbf{X}) = 2m+f(\mathbf{u}))$ on the right-hand side of the above equality varies much more than the factor $P(T_1(\mathbf{X}) \in \mathcal{L}_{m,\mathbf{u}})$. Therefore, the difference between the values $P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u})$ and $P(f(\mathbf{Y}) = m', \mathbf{U} = \mathbf{u}')$ is dictated by the difference between $P(f(\mathbf{X}) = 2m + f(\mathbf{u}))$ and $P(f(\mathbf{X}) = 2m' + f(\mathbf{u}'))$. That is why, for ease of the following derivations, we assume that the probability $P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u})$ is proportional to the probability $P(f(\mathbf{X}) = 2m + f(\mathbf{u}))$, or in other words

$$P(f(\mathbf{Y}) = m, \mathbf{U} = \mathbf{u}) = cP(f(\mathbf{X}) = 2m + f(\mathbf{u})) \tag{26}$$

for some constant c.

To simplify the notation, denote by $P(m, \mathbf{u})(\ell)$ the probability $P(f(\mathbf{Y}') = m, \mathbf{U}' = \mathbf{u})$ for the signal altered by random LSB embedding of length $\ell \in [0, 0.5)$. Note that $P(m, \mathbf{u})(0)$ is the probability corresponding to the clean signal. Based on the approximation (26), we prove next a result that is crucial for the design of the characteristic function $F(\mathbf{V}(\ell))$.

Proposition 1: For any binary sequence $\mathbf{u} = (u_1, u_2, \ldots, u_k)$, k being an odd integer strictly larger than 1, let $\hat{\mathbf{u}}$ be the sequence obtained from $\mathbf{u}$ by flipping the tth bit, $t = \lfloor k/2 \rfloor + 1$. Then, for any $\ell \in (0, 0.5)$ and $m \geq 0$, the following inequalities hold.

1) If $u_t = 1$, then

$$P(m, \mathbf{u})(\ell) - P\left(m + \frac{k-1}{2}, \hat{\mathbf{u}}\right)(\ell) > 0. \tag{27}$$

2) If $u_t = 0$, then

$$P(-m, \mathbf{u})(\ell) - P\left(-m - \frac{k-1}{2}, \hat{\mathbf{u}}\right)(\ell) > 0. \tag{28}$$

The above inequalities change to equalities if $\ell = 0$.

Proof: Consider first the case of $u_t = 1$. It follows from (3) that

$$\begin{aligned} &P(m, \mathbf{u})(\ell) \\ &\quad = \sum_{\mathbf{v}\in\{0,1\}^k} \ell^{d(\mathbf{u},\mathbf{v})}(1-\ell)^{k-d(\mathbf{u},\mathbf{v})} P(m, \mathbf{v})(0) \\ &\quad = \sum_{\mathbf{v}\in\{0,1\}^k, v_t=1} \ell^{d(\mathbf{u},\mathbf{v})}(1-\ell)^{k-d(\mathbf{u},\mathbf{v})} P(m, \mathbf{v})(0) \\ &\qquad + \sum_{\mathbf{v}\in\{0,1\}^k, v_t=0} \ell^{d(\mathbf{u},\mathbf{v})}(1-\ell)^{k-d(\mathbf{u},\mathbf{v})} P(m, \mathbf{v})(0). \end{aligned} \tag{29}$$

When $\mathbf{v}$ ranges over all k-bit binary sequences with $v_t = 1$, $\hat{\mathbf{v}}$ ranges over all k-bit binary sequences with $v_t = 0$. Consequently

$$\begin{aligned} P(m, \mathbf{u})(\ell) = &\sum_{\mathbf{v}\in\{0,1\}^k, v_t=1} \\ &\Big[\ell^{d(\mathbf{u},\mathbf{v})}(1-\ell)^{k-d(\mathbf{u},\mathbf{v})} P(m, \mathbf{v})(0) + \\ &\ell^{d(\mathbf{u},\hat{\mathbf{v}})}(1-\ell)^{k-d(\mathbf{u},\hat{\mathbf{v}})} P(m, \hat{\mathbf{v}})(0)\Big]. \end{aligned} \tag{30}$$

Applying the above to $P(m + (k-1/2), \hat{\mathbf{u}})(\ell)$ as well, we obtain

$$\begin{aligned} &P(m, \mathbf{u})(\ell) - P\left(m + \frac{k-1}{2}, \hat{\mathbf{u}}\right)(\ell) \\ &= \sum_{\mathbf{v}\in\{0,1\}^k, v_t=1} \Big[\ell^{d(\mathbf{u},\mathbf{v})}(1-\ell)^{k-d(\mathbf{u},\mathbf{v})} P(m, \mathbf{v})(0) \\ &\qquad + \ell^{d(\mathbf{u},\hat{\mathbf{v}})}(1-\ell)^{k-d(\mathbf{u},\hat{\mathbf{v}})} P(m, \hat{\mathbf{v}})(0) \\ &\qquad - \ell^{d(\hat{\mathbf{u}},\mathbf{v})}(1-\ell)^{k-d(\hat{\mathbf{u}},\mathbf{v})} P\left(m + \frac{k-1}{2}, \mathbf{v}\right)(0) \\ &\qquad - \ell^{d(\hat{\mathbf{u}},\hat{\mathbf{v}})}(1-\ell)^{k-d(\hat{\mathbf{u}},\hat{\mathbf{v}})} P\left(m + \frac{k-1}{2}, \hat{\mathbf{v}}\right)(0)\Big]. \end{aligned} \tag{31}$$

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Since $u_t = 1, m \geq 0$, and since $\hat{\mathbf{u}}$ (respectively, $\hat{\mathbf{v}}$) differs from $\mathbf{u}$ (respectively, $\mathbf{v}$) only in the tth bit, it follows that $d(\mathbf{u},\mathbf{v}) = d(\hat{\mathbf{u}},\hat{\mathbf{v}})$ and $d(\mathbf{u},\hat{\mathbf{v}}) = d(\hat{\mathbf{u}},\mathbf{v}) = d(\mathbf{u},\mathbf{v}) + 1$. Thus

$$\begin{aligned} &P(m,\mathbf{u})(\ell) - P\left(m + \frac{k-1}{2}, \hat{\mathbf{u}}\right)(\ell) = \\ &\sum_{\mathbf{v}\in\{0,1\}^k, v_t=1} \\ &\Bigg[\ell^{d(\mathbf{u},\mathbf{v})}(1-\ell)^{k-d(\mathbf{u},\mathbf{v})} \\ &\left(P(m,\mathbf{v})(0) - P\left(m+\frac{k-1}{2},\hat{\mathbf{v}}\right)(0)\right) \\ &+ \ell^{d(\mathbf{u},\mathbf{v})+1}(1-\ell)^{k-1-w(\mathbf{u},\mathbf{v})} \\ &\left(P(m,\hat{\mathbf{v}})(0) - P\left(m+\frac{k-1}{2},\mathbf{v}\right)(0)\right)\Bigg]. \end{aligned} \tag{32}$$

Using the assumption of (26), we have

$$\begin{aligned} &P(m,\mathbf{v})(0) - P\left(m+\frac{k-1}{2},\hat{\mathbf{v}}\right)(0) \\ &\quad = c(P(f(\mathbf{X}) = 2m + f(\mathbf{v})) \\ &\qquad - P(f(\mathbf{X}) = 2m + k - 1 + f(\hat{\mathbf{v}}))). \end{aligned} \tag{33}$$

Since $v_t = 1$ and $\hat{\mathbf{v}}$ and $\mathbf{v}$ differ in the tth bit only, $f(\mathbf{v}) = k - 1 + f(\hat{\mathbf{v}})$. Consequently

$$\begin{aligned} &P(f(\mathbf{X}) = 2m + f(\mathbf{v})) \\ &\qquad - P(f(\mathbf{X}) = 2m + k - 1 + f(\hat{\mathbf{v}})) = 0 \end{aligned} \tag{34}$$

which implies

$$P(m,\mathbf{v})(0) - P\left(m+\frac{k-1}{2},\hat{\mathbf{v}}\right)(0) = 0. \tag{35}$$

Note that the above relation also proves the last claim of Proposition 1.

Similarly, from (26) and $f(\mathbf{v}) = k - 1 + f(\hat{\mathbf{v}})$, we have

$$\begin{aligned} &P(m,\hat{\mathbf{v}})(0) - P\left(m+\frac{k-1}{2},\mathbf{v}\right)(0) \\ &\quad = c(P(f(\mathbf{X}) = 2m + f(\hat{\mathbf{v}})) \\ &\qquad - P(f(\mathbf{X}) = 2m + k - 1 + f(\mathbf{v}))) \\ &\quad = c(P(f(\mathbf{X}) = 2m + f(\hat{\mathbf{v}})) \\ &\qquad - P(f(\mathbf{X}) = 2m + 2k - 2 + f(\hat{\mathbf{v}}))). \end{aligned} \tag{36}$$

It is clear that $2m + f(\hat{\mathbf{v}}) < 2m + 2k - 2 + f(\hat{\mathbf{v}})$. When $2m + f(\hat{\mathbf{v}}) \geq 0$, since $f(\mathbf{X})$ is a generalized Gaussian random variable of 0 mean, it follows that $P(f(\mathbf{X}) = 2m + f(\hat{\mathbf{v}})) > P(f(\mathbf{X}) = 2m + 2k - 2 + f(\hat{\mathbf{v}}))$. If $2m + f(\hat{\mathbf{v}}) < 0$, then $P(f(\mathbf{X}) = 2m + f(\hat{\mathbf{v}})) = P(f(\mathbf{X}) = -2m - f(\hat{\mathbf{v}}))$ due to the symmetry of the probability $P(f(\mathbf{X}))$. However, $-2m - f(\hat{\mathbf{v}}) \leq 2m + 2k - 2 + f(\hat{\mathbf{v}})$. Indeed, this inequality is equivalent to $0 \leq 2m + k - 1 + f(\hat{\mathbf{v}})$, which is valid because $m \geq 0$, and $f(\hat{\mathbf{v}}) \geq -k + 1$. This further implies that $P(f(\mathbf{X}) = -2m - f(\hat{\mathbf{v}})) \geq P(f(\mathbf{X}) = 2m + 2k - 2 + f(\hat{\mathbf{v}}))$, where the equality holds only if $2m + k - 1 + f(\hat{\mathbf{v}}) = 0$, which is further true only if $m = 0$ and $f(\hat{\mathbf{v}}) = -k + 1$, i.e., when the binary sequence $\mathbf{v}$ has all the bits equal to one. Consequently

$$P(m,\hat{\mathbf{v}})(0) - P\left(m+\frac{k-1}{2},\mathbf{v}\right)(0) \geq 0 \tag{37}$$

and the inequality is strict if $\mathbf{v}$ has at least one 0 bit. Equations (32), (35), and (37) conclude the proposition for the case $u_t = 1$.

The case $u_t = 0$ can be proved by a similar argument. □

IV. Characteristic Function

After selecting a feature vector $\mathbf{V}$, the final step to arrive at the detection equation of (10) is to construct the characteristic function $F(\mathbf{V})$. Recall that the properties this function has to satisfy are $F(\mathbf{V}(0)) = F_0$ for some signal-independent constant F_0 and $F(\mathbf{V}(\ell)) \neq F_0$ for all $\ell \in (0, 0.5)$.

Proposition 1 reveals important properties of the feature vector $\mathbf{V}$ for odd k, which are now used to design the characteristic function. To make clearer the connection, we express these properties in a more concise form. Consider the vector $\mathbf{W}(\ell)$, whose components are $P(m,\mathbf{u})(\ell) - P(m + (k-1/2), \hat{\mathbf{u}})(\ell)$ for all integers $m \geq 0$ and $\mathbf{u} \in \{0,1\}^k$ such that $u_t = 1$, and $P(-m,\mathbf{u})(\ell) - P(-m - (k-1/2), \hat{\mathbf{u}})(\ell)$ for all integers $m \geq 0$ and $\mathbf{u} \in \{0,1\}^k$ such that $u_t = 0$. Then, one can easily see that Proposition 1 implies

$$\mathbf{W}(0) \equiv \mathbf{0}; \quad \mathbf{W}(\ell) > \mathbf{0}, \ \ \ell \in (0, 0.5) \tag{38}$$

where $\mathbf{0}$ is the vector of all zero elements. The inequality $>$ in (38) should be understood as being satisfied for all components. It follows from (38) that any function of the vector $\mathbf{W}$, which takes the value 0 when all vector components are 0, and which is positive when all vector components are positive, will satisfy the requirements of the characteristic function. Many examples for such a function can be easily found.

However, the above observation has to be used with care. This is because the result of Proposition 1 was derived based on the approximation in (26). Errors in the approximation, due to large difference between $P(T_1(\mathbf{X}) \in \mathcal{L}_{m,\mathbf{u}})$ and $P(T_1(\mathbf{X}) \in \mathcal{L}_{m',\mathbf{u}'})$ when $(m,\mathbf{u}) \neq (m',\mathbf{u}')$, will affect (38).

Next, we derive a more robust characteristic function $F(\mathbf{V}(\ell))$ for odd k. To prove $F(\mathbf{V}(\ell)) > 0$ for $\ell \in (0, 0.5)$, we will rely on Proposition 1, but in order to reduce the sensitivity of $F(\mathbf{V}(\ell))$ to errors in (26), we construct an F such that $F(\mathbf{V}(0)) = F_0$ does not depend on (26).

Let $s(\mathbf{X}) = x_1 + x_2 + \cdots + x_k$, and by the definition of $f(\mathbf{X})$

$$\begin{aligned} s(\mathbf{X}) &= kX_t - f(\mathbf{X}) \\ &= k(2Y_t + U_t) - f(\mathbf{X} = 2kY_t + kU_t - f(\mathbf{X}). \end{aligned} \tag{39}$$

The parity of $s(\mathbf{X})$ depends on bit U_t and the parity of $f(\mathbf{X})$, as given by Table I. Since $f(\mathbf{X})$ is highpass, whereas $s(\mathbf{X})$ is

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TABLE I
PARITY OF $s(\mathbf{X})$ FOR DIFFERENT PARITIES OF $f(\mathbf{X})$ AND U_l

$f(\mathbf{X})$	U_l	$s(\mathbf{X})$
even	0	even
even	1	odd
odd	0	odd
odd	1	even

lowpass, the two random variables can be considered to be independent. This means that $P(s(\mathbf{X}) = \text{even}|f(\mathbf{X}) = n)$, respectively, $P(s(\mathbf{X}) = \text{odd}|f(\mathbf{X}) = n)$, does not depend on the integer n; hence

$$\begin{aligned} &P(s(\mathbf{X}) = \text{odd}|f(\mathbf{X}) = n) - P(s(\mathbf{X}) = \text{even}|f(\mathbf{X}) = n) \\ &\quad = P(s(\mathbf{X}) = \text{odd}|f(\mathbf{X}) = -n) \\ &\qquad - P(s(\mathbf{X}) = \text{even}|f(\mathbf{X}) = -n) \end{aligned} \tag{40}$$

and equivalently

$$\begin{aligned} &\frac{P(f(\mathbf{X}) = n, s(\mathbf{X}) = \text{odd})}{P(f(\mathbf{X}) = n)} - \frac{P(f(\mathbf{X}) = n, s(\mathbf{X}) = \text{even})}{P(f(\mathbf{X}) = n)} \\ &\quad = \frac{P(f(\mathbf{X}) = -n, s(\mathbf{X}) = \text{odd})}{P(f(\mathbf{X}) = -n)} \\ &\qquad - \frac{P(f(\mathbf{X}) = -n, s(\mathbf{X}) = \text{even})}{P(f(\mathbf{X}) = -n)}. \end{aligned} \tag{41}$$

Given that $f(\mathbf{X})$ is generalized Gaussian random variable of zero mean, $P(f(\mathbf{X}) = n) = P(f(\mathbf{X}) = -n)$. In addition, since $s(\mathbf{X}) = 2s(\mathbf{Y}) + s(\mathbf{U})$, $s(\mathbf{X})$ is even/odd if and only if $s(\mathbf{U})$ is even/odd. These considerations, together with (41), lead to

$$\begin{aligned} &P(f(\mathbf{X}) = n, s(\mathbf{U}) = \text{odd}) - P(f(\mathbf{X}) = n, s(\mathbf{U}) = \text{even}) \\ &\quad = P(f(\mathbf{X}) = -n, s(\mathbf{U}) = \text{odd}) \\ &\qquad - P(f(\mathbf{X}) = -n, s(\mathbf{U}) = \text{even}). \end{aligned} \tag{42}$$

Summing up (42) over different values of n yields

$$\begin{aligned} &\sum_{n \geq k-1, n=\text{even}} (P(f(\mathbf{X}) = n, s(\mathbf{U}) = \text{odd}) \\ &- P(f(\mathbf{X}) = n, s(\mathbf{U}) = \text{even})) \\ &+ \sum_{n \geq k-2, n=\text{odd}} (P(f(\mathbf{X}) = n, s(\mathbf{U}) = \text{even}) \\ &- P(f(\mathbf{X}) = n, s(\mathbf{U}) = \text{odd})) \\ &= \sum_{n \geq k-1, n=\text{even}} (P(f(\mathbf{X}) = -n, s(\mathbf{U}) = \text{odd}) \\ &- P(f(\mathbf{X}) = -n, s(\mathbf{U}) = \text{even})) \\ &\quad + \sum_{n \geq k-2, n=\text{odd}} (P(f(\mathbf{X}) = -n, s(\mathbf{U}) = \text{even}) \\ &- P(f(\mathbf{X}) = -n, s(\mathbf{U}) = \text{odd})). \end{aligned} \tag{43}$$

For each pair of integers b, j, with $b \in \{0, 1\}$ and $0 \leq j \leq k-1$, denote by $\mathcal{U}_{b,j}$ the set of k-tuples of bits $\mathbf{u} = (u_1, u_2, \ldots, u_k)$ such that $u_t = b$, and the sum of the other bits equals j. By the same argument following (39), we can show that if $f(\mathbf{X})$ is even, then $s(\mathbf{U})$ is even or odd, depending on if $\mathbf{U} \in \cup_{j=\text{even}} \mathcal{U}_{0,j}$ or $\mathbf{U} \in \cup_{j=\text{even}} \mathcal{U}_{1,j}$. Symmetrically, if $f(\mathbf{X})$ is odd, then $s(\mathbf{U})$ is even or odd, depending on if $\mathbf{U} \in \cup_{j=\text{odd}} \mathcal{U}_{1,j}$ or $\mathbf{U} \in \cup_{j=\text{odd}} \mathcal{U}_{0,j}$. Note further that if $f(\mathbf{X}) = n$ and $\mathbf{U} \in \mathcal{U}_{b,j}$, then $f(\mathbf{U}) = (k-1)b - j$, and $f(\mathbf{Y}) = (n - (k-1)b + j/2)$. Based on the above observations, we conclude the following.

1) For even n (positive or negative)

$$\begin{aligned} &P(f(\mathbf{X}) = n, s(\mathbf{U}) = \text{odd}) \\ &\quad = \sum_{j=0, j=\text{even}}^{k-1} P\left(f(\mathbf{Y}) = \frac{n-k+1+j}{2}, \mathbf{U} \in \mathcal{U}_{1,j}\right); \end{aligned} \tag{44}$$

$$\begin{aligned} &P(f(\mathbf{X}) = n, s(\mathbf{U}) = \text{even}) \\ &\quad = \sum_{j=0, j=\text{even}}^{k-1} P\left(f(\mathbf{Y}) = \frac{n+j}{2}, \mathbf{U} \in \mathcal{U}_{0,j}\right). \end{aligned} \tag{45}$$

2) For odd n (positive or negative)

$$\begin{aligned} &P(f(\mathbf{X}) = n, s(\mathbf{U}) = \text{odd}) \\ &\quad = \sum_{j=1, j=\text{odd}}^{k-2} P\left(f(\mathbf{Y}) = \frac{n+j}{2}, \mathbf{U} \in \mathcal{U}_{0,j}\right); \end{aligned} \tag{46}$$

$$\begin{aligned} &P(f(\mathbf{X}) = n, s(\mathbf{U}) = \text{even}) \\ &\quad = \sum_{j=1, j=\text{odd}}^{k-2} P\left(f(\mathbf{Y}) = \frac{n-k+1+j}{2}, \mathbf{U} \in \mathcal{U}_{1,j}\right). \end{aligned} \tag{47}$$

Let $P(m, b; j)$ be a shorthand notation of $P(f(\mathbf{Y}) = m, \mathbf{U} \in \mathcal{U}_{b,j})$ and rewrite (43) as

$$\begin{aligned} &\sum_{n \geq k-1, n=\text{even}} \sum_{0 \leq j \leq k-1, j=\text{even}} P\left(\frac{n-k+1+j}{2}, 1; j\right) \\ &- \sum_{n \geq k-1, n=\text{even}} \sum_{0 \leq j \leq k-1, j=\text{even}} P\left(\frac{n+j}{2}, 0; j\right) \\ &+ \sum_{n \geq k-2, n=\text{odd}} \sum_{1 \leq j \leq k-2, j=\text{odd}} P\left(\frac{n-k+1+j}{2}, 1; j\right) \\ &- \sum_{n \geq k-2, n=\text{odd}} \sum_{1 \leq j \leq k-2, j=\text{odd}} P\left(\frac{n+j}{2}, 0; j\right) \\ &= \sum_{n \geq k-1, n=\text{even}} \sum_{0 \leq j \leq k-1, j=\text{even}} P\left(\frac{-n-k+1+j}{2}, 1; j\right) \\ &- \sum_{n \geq k-1, n=\text{even}} \sum_{0 \leq j \leq k-1, j=\text{even}} P\left(\frac{-n+j}{2}, 0; j\right) \\ &+ \sum_{n \geq k-2, n=\text{odd}} \sum_{1 \leq j \leq k-2, j=\text{odd}} P\left(\frac{-n-k+1+j}{2}, 1; j\right) \\ &- \sum_{n \geq k-2, n=\text{odd}} \sum_{1 \leq j \leq k-2, j=\text{odd}} P\left(\frac{-n+j}{2}, 0; j\right). \end{aligned} \tag{48}$$

In each of the above eight double summations, we do the change of variables $(n, j) \to (m, j)$ such that $m = (n - k + 1 + j/2)$

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for the sums in the left side, respectively, $m = (n - j/2)$ for the sums in the right side and obtain

$$\begin{aligned}
&\sum_{m\geq 0}\sum_{0\leq j\leq \min(2m,k-1),j=\text{even}} P(m,1;j)\\
&-\sum_{m\geq 0}\sum_{0\leq j\leq \min(2m,k-1),j=\text{even}} P\left(m+\frac{k-1}{2},0;j\right)\\
&+\sum_{m\geq 0}\sum_{1\leq j\leq \min(2m+1,k-2),j=\text{odd}} P(m,1;j)\\
&-\sum_{m\geq 0}\sum_{1\leq j\leq \min(2m+1,k-2),j=\text{odd}} P\left(m+\frac{k-1}{2},0;j\right)\\
&=\sum_{m\geq 0}\sum_{\max(0,k-1-2m)\leq j\leq k-1,j=\text{even}} P\left(-m-\frac{k-1}{2},1;j\right)\\
&-\sum_{m\geq 0}\sum_{\max(0,k-1-2m)\leq j\leq k-1,j=\text{even}} P(-m,0;j)\\
&+\sum_{m\geq 0}\sum_{\max(1,k-2-2m)\leq j\leq k-2,j=\text{odd}} P\left(-m-\frac{k-1}{2},1;j\right)\\
&-\sum_{m\geq 0}\sum_{\max(1,k-2-2m)\leq j\leq k-2,j=\text{odd}} P(-m,0;j)
\end{aligned} \tag{49}$$

which is equivalent to

$$\begin{aligned}
&\sum_{m\geq 0}\sum_{j=0}^{\min(2m+1,k-1)}\\
&\left(P(m,1;j)-P\left(m+\frac{k-1}{2},0;j\right)\right)\\
&=\sum_{m\geq 0}\sum_{j=\max(0,k-2-2m)}^{k-1}\\
&\left(P\left(-m-\frac{k-1}{2},1;j\right)-P(-m,0;j)\right)
\end{aligned} \tag{50}$$

and further equivalent to

$$\begin{aligned}
&\sum_{m\geq 0}\sum_{j=0}^{\min(2m+1,k-1)}\\
&\left(P(m,1;j)-P\left(m+\frac{k-1}{2},0;j\right)\right)\\
&+\sum_{m\geq 0}\sum_{j=\max(0,k-2-2m)}^{k-1}\\
&\left(P(-m,0;j)-P\left(-m-\frac{k-1}{2},1;j\right)\right)=0.
\end{aligned} \tag{51}$$

Recall that

$$P(m,b;j)=\sum_{\mathbf{u}\in\mathcal{U}_{b,j}} P(m,\mathbf{u})(0). \tag{52}$$

As in the previous section, given a sequence $\mathbf{u}$ of k bits, we denote by $\hat{\mathbf{u}}$ the sequence obtained from $\mathbf{u}$ by flipping the tth bit. If $\mathbf{u}$ varies in the set $\mathcal{U}_{b,j}$, then $\hat{\mathbf{u}}$ varies in the set $\mathcal{U}_{1-b,j}$, and thus

$$\begin{aligned}
&P(m,1;j)-P\left(m+\frac{k-1}{2},0;j\right)\\
&\quad=\sum_{\mathbf{u}\in\mathcal{U}_{1,j}}\left(P(m,\mathbf{u})(0)-P\left(m+\frac{k-1}{2},\hat{\mathbf{u}}\right)(0)\right);\\
&P(-m,0;j)-P\left(-m-\frac{k-1}{2},1;j\right)\\
&\quad=\sum_{\mathbf{u}\in\mathcal{U}_{0,j}}\left(P(-m,\mathbf{u})(0)-P\left(-m-\frac{k-1}{2},\hat{\mathbf{u}}\right)(0)\right).
\end{aligned} \tag{53}$$

Substituting (53) into (51), we get

$$\begin{aligned}
&\sum_{m\geq 0}\sum_{j=0}^{\min(2m+1,k-1)}\sum_{\mathbf{u}\in\mathcal{U}_{1,j}}\\
&\left(P(m,\mathbf{u})(0)-P\left(m+\frac{k-1}{2},\hat{\mathbf{u}}\right)(0)\right)\\
&+\sum_{m\geq 0}\sum_{j=\max(0,k-2-2m)}^{k-1}\sum_{\mathbf{u}\in\mathcal{U}_{0,j}}\\
&\left(P(-m,\mathbf{u})(0)-P\left(-m-\frac{k-1}{2},\hat{\mathbf{u}}\right)(0)\right)=0.
\end{aligned} \tag{54}$$

Finally, we define the characteristic function $F(\mathbf{V}(\ell))$:

$$\begin{aligned}
F(\mathbf{V}(\ell))=&\sum_{m\geq 0}\sum_{j=0}^{\min(2m+1,k-1)}\sum_{\mathbf{u}\in\mathcal{U}_{1,j}}\\
&\left(P(m,\mathbf{u})(\ell)-P\left(m+\frac{k-1}{2},\hat{\mathbf{u}}\right)(\ell)\right)\\
&+\sum_{m\geq 0}\sum_{j=\max(0,k-2-2m)}^{k-1}\sum_{\mathbf{u}\in\mathcal{U}_{0,j}}\\
&\left(P(-m,\mathbf{u})(\ell)-P\left(-m-\frac{k-1}{2},\hat{\mathbf{u}}\right)(\ell)\right).
\end{aligned} \tag{55}$$

It is easy to see that $F(\mathbf{V}(0)) = 0$ by (54). Furthermore, by Proposition 1, we have $F(\mathbf{V}(\ell)) > 0$ for $\ell \in (0, 0.5)$. Consequently, $F(\mathbf{V}(\ell))$ satisfies the requirements of a characteristic function.

Referring back to the discussion in Section II, given the characteristic function $F(\mathbf{V})$, the embedded message length in the tested signal can be estimated by finding the smallest non-negative solution of the equation

$$F(\bar{\mathbf{Q}}^{-1}(\ell)\mathbf{V}')=0 \tag{56}$$

where $\mathbf{V}'$ denotes the value of the vector $\mathbf{V}$ for the tested signal. In order to derive the explicit form of the detection equation (56), for the characteristic function F developed above, each quantity $P(\cdot,\cdot)(0)$ in (54) has to be expressed using (6) in terms of ℓ and quantities $P'(m,\mathbf{u}')$, where $P'(m,\mathbf{u}')$ is the probability $P(f(\mathbf{Y}') = m, \mathbf{U}' = \mathbf{u}')$ for the tested signal.

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Using the result $\mathbf{Q}^{-1}(\ell) = \mathbf{Q}(-\ell/1-2\ell)$, which is proven in Appendix A, we have

$$\begin{aligned} P(m,\mathbf{u})(0) &= \sum_{\mathbf{v}\in\{0,1\}^k} \left(\frac{-\ell}{1-2\ell}\right)^{d(\mathbf{u},\mathbf{v})} \\ &\quad \times \left(1-\frac{-\ell}{1-2\ell}\right)^{k-d(\mathbf{u},\mathbf{v}))} P'(m,\mathbf{v}) \\ &= \frac{1}{(1-2\ell)^k} \sum_{\mathbf{v}\in\{0,1\}^k} (-\ell)^{w(\mathbf{u},\mathbf{v})} \\ &\quad \times (1-\ell)^{k-d(\mathbf{u},\mathbf{v})} P'(m,\mathbf{v}). \end{aligned} \tag{57}$$

An explicit form of detection equation (56) can be obtained by substituting (57) into (54) and multiplying both sides by $(1-2\ell)^k$. Note that the equation obtained is a polynomial of order k.

V. Case Study for $k = 3$

In this section, we exemplify the application of the proposed LSB steganlysis framework by deriving the explicit form of the detection equation for the case $k = 3$. This detection equation turns out to be a cubic polynomial whose coefficients can be determined by the sample statistics of the signal being examined only, without any knowledge of the original signal. This cubic detection equation offers an efficient steganalysis procedure that exploits the correlation among three consecutive samples.

In order to obtain a simpler form for the detection equation, we do not substitute (57) directly into (54) but use an equivalent form of (54). As shown in Appendix B, (55) is equivalent to

$$\begin{aligned} &\sum_{m=0}^{\frac{k-3}{2}} \sum_{j=0}^{2m+1} (P(m,1;j) + P(-m,0;k-1-j)) \\ &+ \sum_{m=(k-1)/2}^{k-2} \sum_{j=2m-k+3}^{k-1} (P(m,0;j) \\ &+ P(-m,1;k-1-j)) \\ &\quad + \sum_{m\geq\frac{k-1}{2}} (P(m,1) - P(m,0) \\ &+ P(-m,0) - P(-m,1)) = 0 \end{aligned} \tag{58}$$

where $P(m,b)$ denotes the probability $P(f(\mathbf{Y}) = m, U_t = b)$ for the clean signal, for any integers m and b with $b \in \{0,1\}$. Later, we will also use the notation $P'(m,b)$ for the corresponding probability measured on the tested signal.

By letting $k = 3$ in (58), we obtain

$$\begin{aligned} &\sum_{j=0}^{1}(P(0,1;j) + P(0,0;2-j)) + P(1,0;2) + P(-1,1;0) \\ &\quad + \sum_{m\geq 1}(P(m,1) - P(m,0) + P(-m,0) \\ &\qquad - P(-m,1)) = 0. \end{aligned} \tag{59}$$

We denote $P(m) = P(f(\mathbf{Y}) = m)$ for any integer m. Accordingly, $P'(m)$ is defined for the tested signal. Since $P(0) = \sum_{j=0}^{2}(P(0,0;j) + P(0,1;j))$, we can rewrite (59) as

$$\begin{aligned} &P(0) - P(0,1;2) - P(0,0;0) + P(1,0;2) + P(-1,1;0) \\ &\quad + \sum_{m\geq 1}(P(m,1) - P(m,0) + P(-m,0) \\ &\qquad - P(-m,1)) = 0. \end{aligned} \tag{60}$$

For simplicity, we use, in the sequel, the notation $P(m,\mathbf{u})$ instead of $P(m,\mathbf{u})(0)$. By the definition of $P(m,b;j) = P(f(\mathbf{Y}) = m, \mathbf{U} \in \mathcal{U}_{b,j})$, we have

$$\begin{aligned} P(0,1;2) &= P(0{,}111) \\ P(0,0;0) &= P(0{,}000) \\ P(1,0;2) &= P(1{,}101) \\ P(-1,1;0) &= P(-1{,}010). \end{aligned} \tag{61}$$

Thus, (60) becomes

$$\begin{aligned} &P(0) - P(0{,}111) - P(0{,}000) + P(1{,}101) + P(-1{,}010) \\ &\quad + \sum_{m\geq 1}\Big[P(m,1) - P(m,0) + P(-m,0) \\ &\qquad - P(-m,1)\Big] = 0. \end{aligned} \tag{62}$$

Next, we express the terms of (62) in observable quantities of the signal being steganalyzed and the hidden message length ℓ, using (57). Let $\hat{\mathbf{v}}$ be the bitwise inverse of the binary sequence $\mathbf{v}$. Since $d(000,\mathbf{v}) = w(\mathbf{v})$ and $d(111,\tilde{\mathbf{v}}) = w(\mathbf{v})$, where $w(\mathbf{v})$ denotes the Hamming weight of the binary vector $\mathbf{v}$, we have

$$\begin{aligned} &P(0{,}000) + P(0{,}111) \\ &\quad = \frac{1}{(1-2\ell)^3} \sum_{\mathbf{v}\in\{0,1\}^3} (-\ell)^{w(\mathbf{v})}(1-\ell)^{3-w(\mathbf{v})} \\ &\qquad \times [P'(0,\mathbf{v}) + P'(0,\tilde{\mathbf{v}})] \\ &\quad = \frac{1}{(1-2\ell)^3} \sum_{d=0}^{3}(-\ell)^d(1-\ell)^{3-d} \\ &\qquad \times \sum_{\mathbf{v}\in\{0,1\}^3, w(\mathbf{v})=d} [P'(0,\mathbf{v}) + P'(0,\tilde{\mathbf{v}})]. \end{aligned} \tag{63}$$

It is easy to show that

$$\begin{aligned} &\sum_{\mathbf{v}\in\{0,1\}^3, w(\mathbf{v})=1} [P'(0,\mathbf{v}) + P'(0,\tilde{\mathbf{v}})] \\ &\quad = \sum_{\mathbf{v}\in\{0,1\}^3, w(\mathbf{v})=2} [P'(0,\mathbf{v}) + P'(0,\tilde{\mathbf{v}})] \\ &\quad = P'(0) - P'(0{,}000) - P'(0{,}111). \end{aligned} \tag{64}$$

Replacing in (63) and via straightforward computations, we arrive at

$$\begin{aligned} &P(0{,}000) + P(0{,}111) \\ &= \frac{1}{(1-2\ell)^2}[P'(0{,}000) + P'(0{,}111) - \ell(1-\ell)P'(0)]. \end{aligned} \tag{65}$$

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Note that $P(0) = P'(0)$ because embedding does not affect the most significant $B-1$ bits. Then

$$P(0) - P(0{,}000) - P(0{,}111) = \frac{(1-3\ell+3\ell^2)P'(0) - P'(0{,}000) - P'(0{,}111)}{(1-2\ell)^2}. \quad (66)$$

By applying (57), we also obtain

$$\begin{aligned} &P(1{,}101) \\ &= \frac{1}{(1-2\ell)^3}[(1-\ell)^3 P'(1{,}101) \\ &- \ell(1-\ell)^2[P'(1{,}001) + P'(1{,}100) + P'(1{,}111)] \\ &+ \ell^2(1-\ell)[P'(1{,}000) + P'(1{,}011) + P'(1{,}110)] \\ &- \ell^3 P'(1{,}010)]; \\ &P(-1{,}010) \\ &= \frac{1}{(1-2\ell)^3}[(1-\ell)^3 P'(-1{,}010) \\ &- \ell(1-\ell)^2[P'(-1{,}000) + P'(-1{,}011) + P'(-1{,}110)] \\ &+ \ell^2(1-\ell)[P'(-1{,}001) + P'(-1{,}100) \\ &+ P'(-1{,}111)] - \ell^3 P'(-1{,}101)]. \end{aligned} \quad (67)$$

To express the values $P(m,b)$ of (58) in terms of quantities measurable on the tested signal, we can use a relation much simpler than (57). Note that

$$\begin{aligned} &P(f(\mathbf{Y}') = m, U'_t = b) \\ &= P(f(\mathbf{Y}) = m, U_t = b)P(U'_t = b|U_t = b) \\ &\quad + P(f(\mathbf{Y}) = m, U_t = 1-b)P(U'_t = 1-b|U_t = b) \\ &= (1-\ell)P(f(\mathbf{Y}) = m, U_t = b) \\ &\quad + \ell P(f(\mathbf{Y}) = m, U_t = 1-b). \end{aligned} \quad (68)$$

The above relation implies

$$\begin{aligned} P'(m,1) - P'(m,0) &= (1-\ell)P(m,1) + \ell P(m,0) \\ &\quad - (1-\ell)P(m,0) - \ell P(m,1) \\ &= (1-2\ell)(P(m,1) - P(m,0)) \end{aligned} \quad (69)$$

which leads further to

$$P(m,1) - P(m,0) = \frac{1}{1-2\ell}[P'(m,1) - P'(m,0)]. \quad (70)$$

Consequently, the last sum in equality (58) can be expressed as

$$\begin{aligned} &\sum_{m\geq 1}[P(m,1) - P(m,0) + P(-m,0) - P(-m,1)] \\ &= \frac{1}{1-2\ell}\sum_{m\geq 1}\Big[P'(m,1) - P'(m,0) \\ &\qquad + P'(-m,0) - P'(-m,1)\Big] \\ &= \frac{1}{1-2\ell}\sum_{m\geq 1}\Big[P'(m) - 2P'(m,0) \\ &\qquad + 2P'(-m,0) - P'(-m)\Big]. \end{aligned} \quad (71)$$

The last step of (71) follows from the relation $P'(m,1) = P'(m) - P'(m,0)$.

Putting (63) through (71) together, we can express the detection equation (62) as

$$\begin{aligned} &(1-2\ell)(1-3\ell+3\ell^2)C_0 - (1-2\ell)C_1 + (1-\ell)^3 C_2 \\ &-\ell(1-\ell)^2 C_3 + \ell^2(1-\ell)C_4 - \ell^3 C_5 + (1-2\ell)^2 C_6 = 0 \end{aligned} \quad (72)$$

where

$$\begin{aligned} C_0 &= P'(0) \\ C_1 &= P'(0{,}000) + P'(0{,}111) \\ C_2 &= P'(1{,}101) + P'(-1{,}010) \\ C_3 &= P'(1{,}001) + P'(1{,}100) + P'(1{,}111) + P'(-1{,}000) \\ &\quad + P'(-1{,}011) + P'(-1{,}110) \\ C_4 &= P'(1{,}000) + P'(1{,}011) + P'(1{,}110) \\ &\quad + P'(-1{,}001) + P'(-1{,}100) + P'(-1{,}111) \\ C_5 &= P'(1{,}010) + P'(-1{,}101) \\ C_6 &= \sum_{m\geq 1}[P'(m) - 2P'(m,0) + 2P'(-m,0) - P'(-m)]. \end{aligned} \quad (73)$$

Finally, the detection equation can be simplified into the following cubic polynomial in the estimated hidden message length ℓ:

$$\begin{aligned} &(-6C_0 - C_2 - C_3 - C_4 - C_5)\ell^3 \\ &\quad + (9C_0 + 3C_2 + 2C_3 + C_4 + 4C_6)\ell^2 \\ &\quad + (-5C_0 + 2C_1 - 3C_2 - C_3 - 4C_6)\ell \\ &\quad + C_0 - C_1 + C_2 + C_6 = 0. \end{aligned} \quad (74)$$

We implemented the estimation method based on the cubic polynomial equation (74) (we refer to this method as "triple") and compared it with the LSB steganalysis methods of [2] and of [10]. The method of [2] (referred to as the "pair" method) uses a quadratic detection equation that is derived from a statistical analysis of sample pairs. It has a competitive performance compared with the RS method proposed by Fridrich *et al.* [5], which also uses a quadratic detection equation. The method of [10] (called the LSM method by the authors) is a least-square variant of the "pair" method.

We conducted experiments on three sets of test images. The first set consists of 29 continuous-tone gray-scale raw images of size 512×512 or 576×720. This set includes all original ISO/JPEG test images such as barb, balloon, goldhill, girl, etc. This is the same test set used in [2]. The second set was obtained by compressing the images of the first set. The compression method was the JPEG lossy compression standard (in the highest quality setting of Matlab). The third set consists of ten color, very high-resolution (2310×1814), uncompressed scanned images. The second test set was created to evaluate the effects of compression on the different steganalysis methods. The third test set was chosen to empirically examine if and how the estimation accuracy of these methods change as the image resolution increases.

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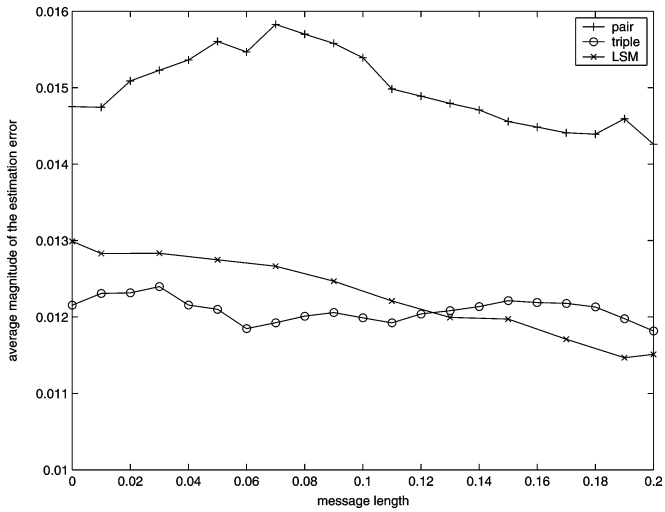


Fig. 1. Average magnitudes of estimation errors versus the true hidden message lengths (2ℓ) for the first test set (typical uncompressed images circulated in the Internet).

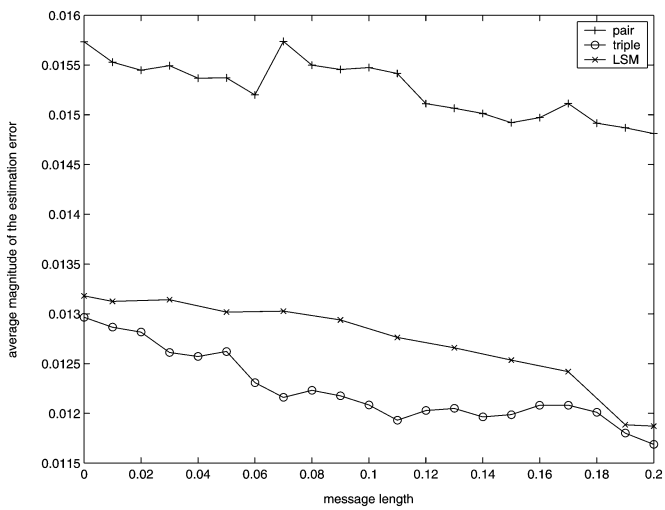
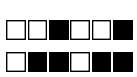


Fig. 2. Average magnitudes of estimation errors versus the true hidden message lengths (2ℓ) for the second test set (the same as the first test set but compressed).

In our implementation of the estimation equation (74), the sample blocks analyzed in the test images consisted of three pixels forming an L-shape:



This design of interlocking two-dimensional sample blocks is to account for sample correlations in both horizontal and vertical directions.

We used the LSB steganalysis software that was gracefully provided to us by the authors of [10] to test the LSM method.

Figs. 1–3 compare the performances of the three LSB steganalysis methods on the three sets of test images, respectively. The three curves of each figure present the average magnitudes of estimation errors for the three LSB steganalysis methods versus the hidden message lengths 2ℓ. For each image and each

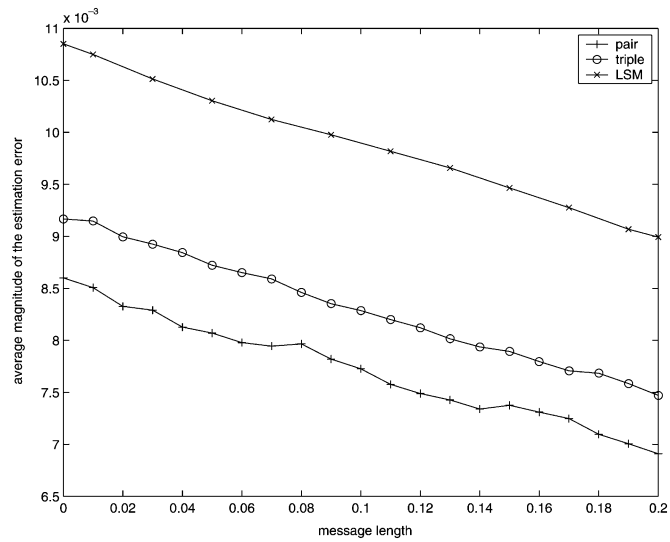


Fig. 3. Average magnitudes of estimation errors versus the true hidden message lengths (2ℓ) for the third test set (high-resolution uncompressed scanned color images).

message length 2ℓ, each estimation method was tested on ten different embedded messages. The average magnitude of the estimation errors was computed over all the test runs performed on each image set, each 2ℓ, and each detection method.

Among the three methods being evaluated, our experiments identified no consistent winner over all test sets. However, the new triple method seemed to be the most robust among the three, considering that it was the best for the first and second test sets and was the second best for the third test set. The proposed triple method obtained more accurate estimates than our previous pair method on images of medium resolutions as those typically encountered in the Internet, raw or compressed, achieving on average 19% lower estimation error. As shown by Fig. 1, the triple method and the LSM method were virtually tied in performance on the first test set, with the former being slightly more accurate than the latter for short hidden message lengths. Fig. 2 demonstrates that the triple method performed better than the LSM method over all message lengths, although the margin was very slim.

All three methods improved their estimation accuracy on high-resolution images (compare Fig. 3 with the other two). This should be expected as sample correlation becomes stronger as image resolution increases, but the relative rate of improvement is quite different for the three methods. The pair method jumped from the worst place to first place, whereas the LSM method improved the least and fell behind the other methods in estimation accuracy. This is somewhat surprising considering that the LSM method was an "optimized" version of the pair method. The triple method held the second place because it was slightly worse than the pair method on the third test set. In summary, our experimental results showed a slight advantage of the new triple method statistically but were still not conclusive enough to answer the question of whether using larger sample blocks in LSB steganalysis will lead to better estimation. Further study is warranted.

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VI. Conclusion

A new general framework for LSB steganalysis of digital media contents was developed. For natural digital signals encountered in daily life such as image, video, and audio, the proposed framework can be used to estimate the hidden message length. A case study on the LSB steganalysis of natural grey-scale and color images was conducted. The experimental results demonstrated the effectiveness and robustness of the new steganalysis framework.

Appendix A

We show here that the matrix equality

$$\mathbf{Q}(\ell_2)\mathbf{Q}(\ell_1) = \mathbf{Q}(\ell_1 + \ell_2 - 2\ell_1\ell_2) \tag{75}$$

holds for any real values ℓ_1, ℓ_2.

A direct proof can be given to these result by evaluating first the product $\mathbf{Q}(\ell_2)\mathbf{Q}(\ell_1)$ and then showing the component wise equality. However, we will present an indirect proof, which is much simpler. Note that a matrix $\mathbf{Q}(\ell)$, where $0 \le \ell \le 1$, can be regarded as the transition matrix of a binary symmetric channel (BSC) with crossover probability ℓ. Moreover, for $0 \le \ell_1, \ell_2 \le 1$, the product $\mathbf{Q}(\ell_2)\mathbf{Q}(\ell_1)$ is actually the transition matrix of a channel obtained by cascading two BSC channels, with crossover probabilities ℓ_1, respectively, ℓ_2. Such a cascaded channel is actually a BSC with crossover probability $\ell_1 + \ell_2 - 2\ell_1\ell_2$. Indeed, a bit is flipped by the cascaded channel if either it is flipped by the first channel and further unchanged by the second or if it is unchanged by the first channel but flipped by the second one. Consequently, the probability that a bit is flipped by the cascaded channel equals $\ell_1(1-\ell_2) + (1-\ell_1)\ell_2 = \ell_1 + \ell_2 - 2\ell_1\ell_2$. Therefore, the transition matrix of the cascaded channel is $\mathbf{Q}(\ell_1+\ell_2-2\ell_1\ell_2)$. Consequently, we have proved the equality (76) for all real values ℓ_1, ℓ_2 in the interval $[0, 1]$. From here, it follows easily that it also holds for all real values ℓ_1, ℓ_2. Indeed, note that each component of the matrix $\mathbf{Q}(\ell_2)\mathbf{Q}(\ell_1)$ and each component of the matrix $\mathbf{Q}(\ell_1 + \ell_2 - 2\ell_1\ell_2)$ have polynomial expressions in each of the variables ℓ_1 and ℓ_2. It is known that if two polynomial functions take equal values on an infinite set (the interval $[0, 1]$ in our case), then they are identical. Thus, the proof is completed.

Equation (75) can be used to find the inverse $\mathbf{Q}^{-1}(\ell)$ of the matrix $\mathbf{Q}(\ell)$. Note that $\mathbf{Q}(0)$ is the identity matrix. Further, $\mathbf{Q}(\ell)\mathbf{Q}(\ell') = \mathbf{Q}(0)$ if and only if $\ell + \ell' - 2\ell\ell' = 0$, i.e., $\ell' = (\frac{-\ell}{1-2\ell})$. Consequently, the matrix $\mathbf{Q}(\ell)$ is invertible only if $\ell \neq 0.5$ and, in this case, $\mathbf{Q}^{-1}(\ell) = \mathbf{Q}(\frac{-\ell}{1-2\ell})$.

Appendix B
Equivalence of (54) and (58)

As shown in Section IV, (54) is equivalent to (51) which can be written as

$$\begin{aligned}
&\sum_{m=0}^{\frac{k-3}{2}} \sum_{j=0}^{2m+1} \left(P(m,1;j) - P\left(m + \frac{k-1}{2}, 0; j\right)\right) \\
&\quad + \sum_{m \ge \frac{k-1}{2}} \sum_{j=0}^{k-1} \left(P(m,1;j) - P\left(m + \frac{k-1}{2}, 0; j\right)\right) \\
&\quad + \sum_{m=0}^{\frac{k-3}{2}} \sum_{j=k-2-2m}^{k-1} \left(P(-m,0;j) - P\left(-m - \frac{k-1}{2}, 1; j\right)\right) \\
&\quad + \sum_{m \ge \frac{k-1}{2}} \sum_{j=0}^{k-1} \left(P(-m,0;j) - P\left(-m - \frac{k-1}{2}, 1; j\right)\right) \\
&= 0.
\end{aligned} \tag{76}$$

Note that $\sum_{j=0}^{k-1} P(m,b;j) = P(f(\mathbf{Y} = m, U_l = b)$. To simplify the notation, we denote this probability by $P(m,b)$. Thus, the above equality becomes

$$\begin{aligned}
&\sum_{m=0}^{\frac{k-3}{2}} \sum_{j=0}^{2m+1} \left(P(m,1;j) - P\left(m + \frac{k-1}{2}, 0; j\right)\right) \\
&\quad + \sum_{m \ge \frac{k-1}{2}} \left(P(m,1) - P\left(m + \frac{k-1}{2}, 0\right)\right) \\
&\quad + \sum_{m=0}^{\frac{k-3}{2}} \sum_{j=k-2-2m}^{k-1} \left(P(-m,0;j) - P\left(-m - \frac{k-1}{2}, 1; j\right)\right) \\
&\quad + \sum_{m \ge \frac{k-1}{2}} \left(P(-m,0) - P\left(-m - \frac{k-1}{2}, 1\right)\right) \\
&= 0.
\end{aligned} \tag{77}$$

By adding and subtracting in the left side the quantities $\sum_{m=0}^{\frac{k-3}{2}} P(m + \frac{k-1}{2}, 0)$ and $\sum_{m=0}^{\frac{k-3}{2}} P(-m - \frac{k-1}{2}, 1)$, we have

$$\begin{aligned}
&\sum_{m=0}^{\frac{k-3}{2}} \left[\sum_{j=0}^{2m+1} \left(P(m,1;j) - P\left(m + \frac{k-1}{2}, 0; j\right)\right) \right. \\
&\left. + P\left(m + \frac{k-1}{2}, 0\right) \right] + \sum_{m \ge \frac{k-1}{2}} \left(P(m,1) - P\left(m + \frac{k-1}{2}, 0\right)\right) \\
&- \sum_{m=0}^{\frac{k-3}{2}} P\left(m + \frac{k-1}{2}, 0\right) \\
&\quad + \sum_{m=0}^{\frac{k-3}{2}} \left[\sum_{j=k-2-2m}^{k-1} \left(P(-m,0;j) - P\left(-m - \frac{k-1}{2}, 1; j\right)\right) \right. \\
&\left. + P\left(-m - \frac{k-1}{2}, 1\right) \right] \\
&\quad + \sum_{m \ge \frac{k-1}{2}} \left(P(-m,0) - P\left(-m - \frac{k-1}{2}, 1\right)\right) \\
&- \sum_{m=0}^{\frac{k-3}{2}} P\left(-m - \frac{k-1}{2}, 1\right) = 0
\end{aligned} \tag{78}$$

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which can be further rewritten as

$$\begin{aligned}&\sum_{m=0}^{\frac{k-3}{2}}\sum_{j=0}^{2m+1}P(m,1;j)+\sum_{m=0}^{\frac{k-3}{2}}\sum_{j=2m+2}^{k-1}\\&\quad\times P\left(m+\frac{k-1}{2},0;j\right)\\&\quad+\sum_{m\geq\frac{k-1}{2}}P(m,1)-\sum_{m\geq 0}P\left(m+\frac{k-1}{2},0\right)\\&\quad+\sum_{m=0}^{\frac{k-3}{2}}\sum_{j=k-2-2m}^{k-1}P(-m,0;j)\\&\quad+\sum_{m=0}^{\frac{k-3}{2}}\sum_{j=0}^{k-3-2m}P\left(-m-\frac{k-1}{2},1;j\right)\\&\quad+\sum_{m\geq\frac{k-1}{2}}P(-m,0)-\sum_{m\geq 0}P\left(-m-\frac{k-1}{2},1\right)=0.\end{aligned}\tag{79}$$

By making the change of variable $m=m+\frac{k-1}{2}$ in the second, fourth, sixth, and eighth sums, we obtain

$$\begin{aligned}&\sum_{m=0}^{\frac{k-3}{2}}\sum_{j=0}^{2m+1}P(m,1;j)+\sum_{m=\frac{k-1}{2}}^{k-2}\sum_{j=2m-k+3}^{k-1}P(m,0;j)\\&\quad+\sum_{m\geq\frac{k-1}{2}}(P(m,1)-P(m,0))\\&\quad+\sum_{m=0}^{\frac{k-3}{2}}\sum_{j=k-2-2m}^{k-1}P(-m,0;j)\\&\quad+\sum_{m=\frac{k-1}{2}}^{k-2}\sum_{j=0}^{2k-4-2m}P(-m,1;j)\\&\quad+\sum_{m\geq\frac{k-1}{2}}(P(-m,0)-P(-m,1))=0\end{aligned}\tag{80}$$

and reordering the terms leads to

$$\begin{aligned}&\sum_{m=0}^{\frac{k-3}{2}}\left(\sum_{j=0}^{2m+1}P(m,1;j)+\sum_{j=k-2-2m}^{k-1}P(-m,0;j)\right)\\&\quad+\sum_{m=\frac{k-1}{2}}^{k-2}\left(\sum_{j=2m-k+3}^{k-1}P(m,0;j)+\sum_{j=0}^{2k-4-2m}P(-m,1;j)\right)\\&\quad+\sum_{m\geq\frac{k-1}{2}}(P(m,1)-P(m,0)\\&\qquad+P(-m,0)-P(-m,1))=0.\end{aligned}\tag{81}$$

Again making a change of variable, $j=k-1-j$ in the second and fourth sums yields (58).

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